Assignment 2 Report

Data Collection and Organizing

1. **How many datapoints do we have for each company a year? Why don't we have 365 datapoints for a year?**

<u>Answer:</u>

The stock market operates only on business days, typically Monday through Friday, excluding public holidays. As a result, there are fewer than 365 data points in a year. For each company, I found 251 data points, which aligns with the typical number of trading days in a year. While a standard year usually has 252 trading days, this number can vary slightly depending on market holidays. The absence of data on weekends and public holidays, such as New Year's Day, Christmas Day, and Thanksgiving in the U.S., explains why we do not have daily stock prices for the entire year (365 days).

2. **What is the data type of stock prices? Cross-section, time series, or panel data?**

<u>Answer:</u>

Stock prices we are working with are considered time series data, as they are recorded at regular intervals, such as daily closing prices. This type of data is essential for analyzing price movements over time, helping to identify trends and patterns in the market.

Statistics

1. **Which of the stocks had the highest annual return and why is this interesting? If you invested $100 last year in this asset, how much will you have in total now?**

<u>Answer:</u>

NVDA had the highest annual return of 91.15%, indicating a significant increase in its value over the past year. This impressive performance suggests that NVDA excelled in the market, potentially due to positive company developments, product innovations, or favorable market sentiment. If I had invested $100 in NVDA last year, that investment would now be worth approximately $191.15. This is calculated by multiplying the initial $100 by the return of 91.15%, demonstrating substantial growth over the period.

2. **Which of the stocks had the highest standard deviation and why is this interesting?**

Answer:

TSLA had the highest standard deviation of 87.14, which shows a high level of volatility in its stock price. This is interesting because it suggests that TSLA's price fluctuates significantly over time, which could mean that it is a riskier investment. The larger the standard deviation, the more unpredictable and volatile the stock's price movements are. Therefore, investors need to consider this when deciding whether or not to invest in TSLA.

3. **Which of the stocks had the lowest standard deviation and why is this interesting?**

Answer:

GM had the lowest standard deviation of 4.60. This is interesting because it indicates that GM's stock price is relatively stable and does not experience large fluctuations. This stability may be appealing to investors who prefer a more predictable investment, as it typically suggests lower risk compared to more volatile stocks. However, the trade-off might be slower growth compared to more volatile assets like TSLA.

4. **Which of the stock pairs had the highest correlation and why is this interesting?**

Answers:

The highest correlation is between NVDA and SPY with a value of 0.90. This is interesting because it means that NVDA and SPY tend to move in the same direction most of the time. A high correlation like this suggests that the performance of NVDA is closely tied to the performance of SPY, which could indicate that both are influenced by similar macroeconomic factors or investor sentiment.

5. **Which of the stock pairs had the lowest correlation and why is this interesting?**

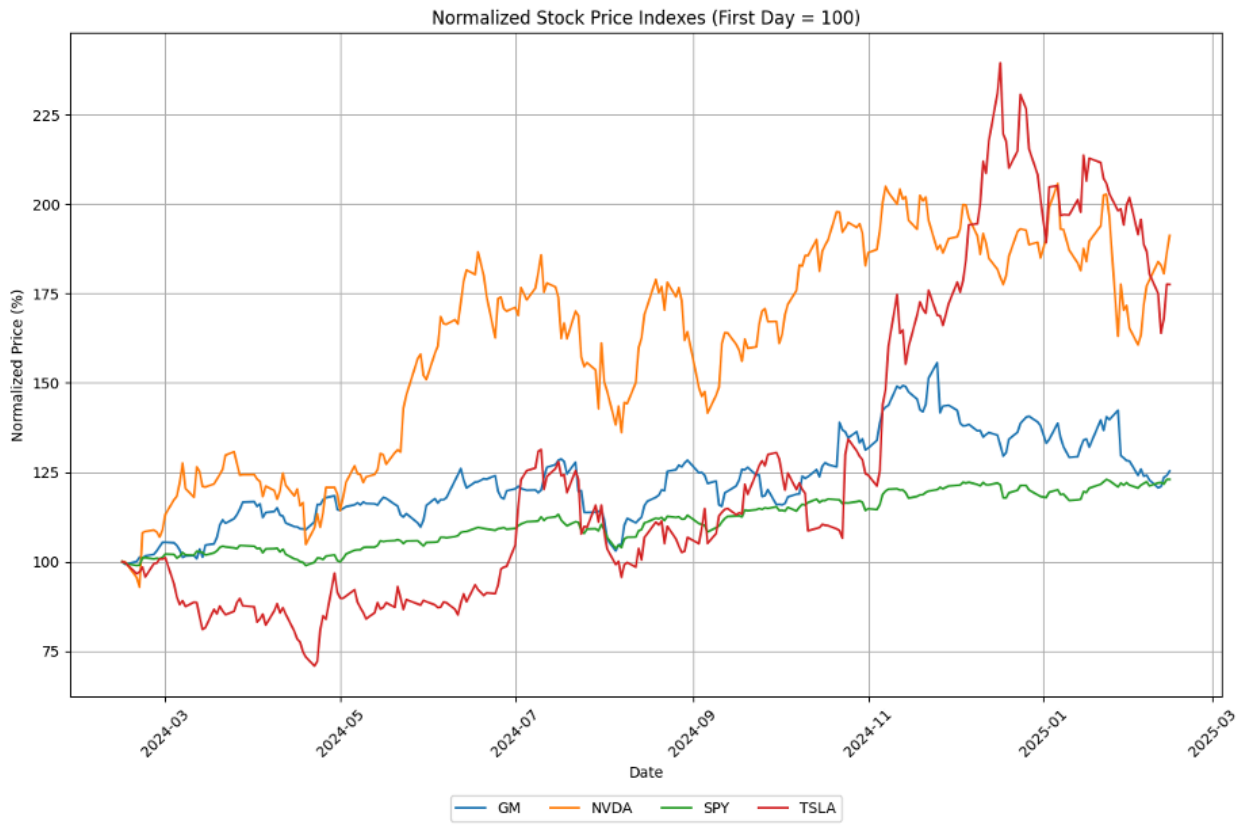
Answers:

The lowest correlation is between TSLA and NVDA with a value of 0.69. This is interesting because it suggests that these two stocks do not move in perfect harmony with each other. While they do have a positive correlation, it is relatively weak compared to other pairs like NVDA and

SPY. A lower correlation can indicate that the stocks might be influenced by different factors, and therefore, holding both could provide some diversification in a portfolio.

Data Visualization

A. Stock Price Index

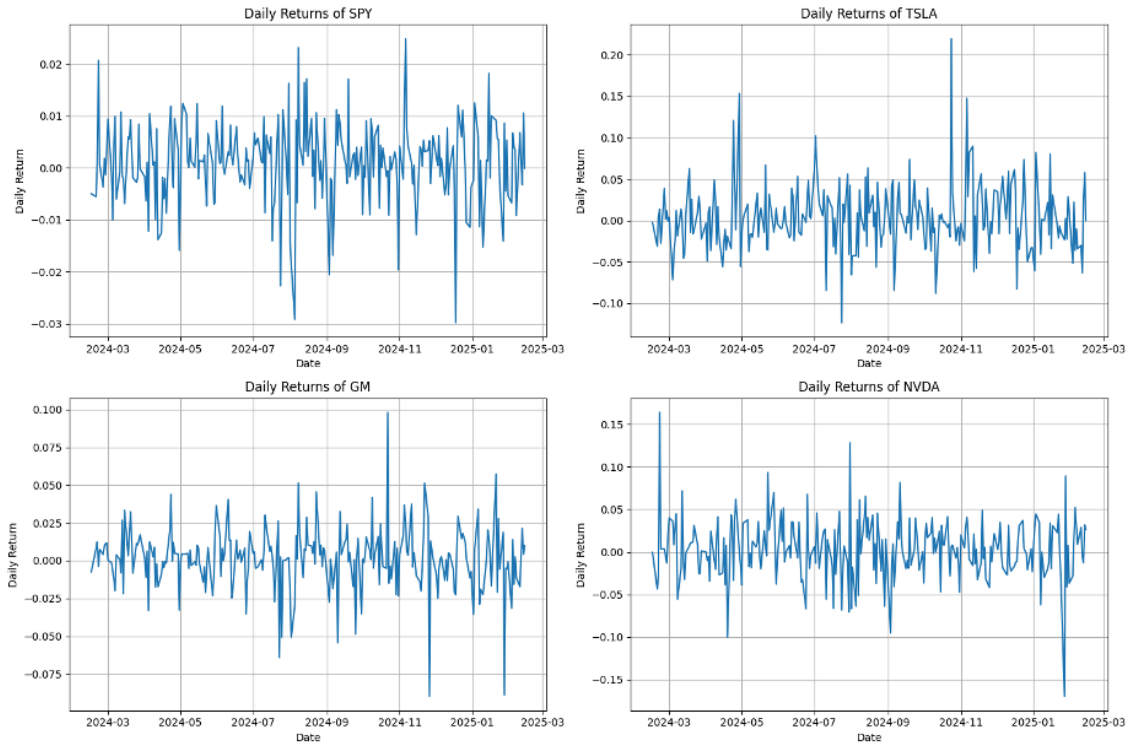


1. **According to your "Close 100" chart, which stock outperforms the market?**

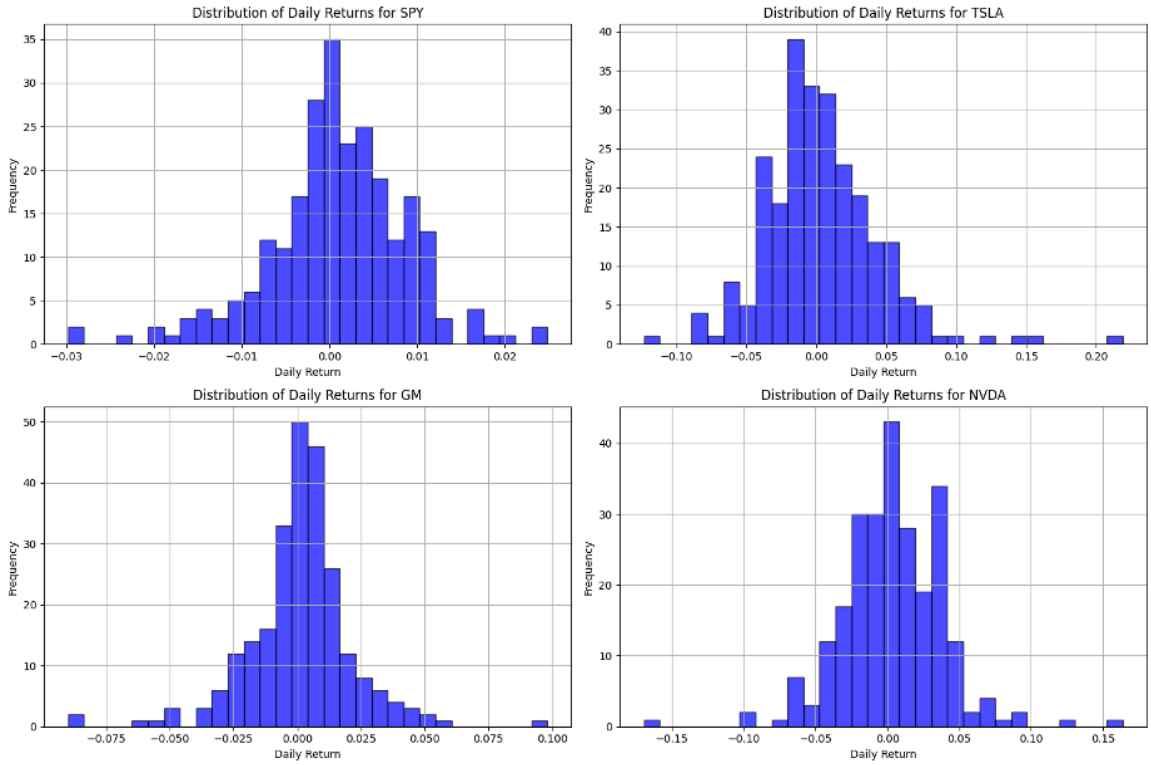
<u>Answer:</u>

Based on the "Close 100" chart, the stocks that outperformed the market (represented by SPY) are NVIDIA (NVDA) and Tesla (TSLA). Their normalized prices increased significantly over time, reaching much higher values compared to SPY. NVIDIA, in particular, showed a strong upward trend throughout the period, while Tesla exhibited more volatility but still managed to surpass the market index. In contrast, General Motors (GM) followed a steady but moderate increase, staying relatively close to SPY. This indicates that NVDA and TSLA delivered stronger returns compared to the overall market performance.

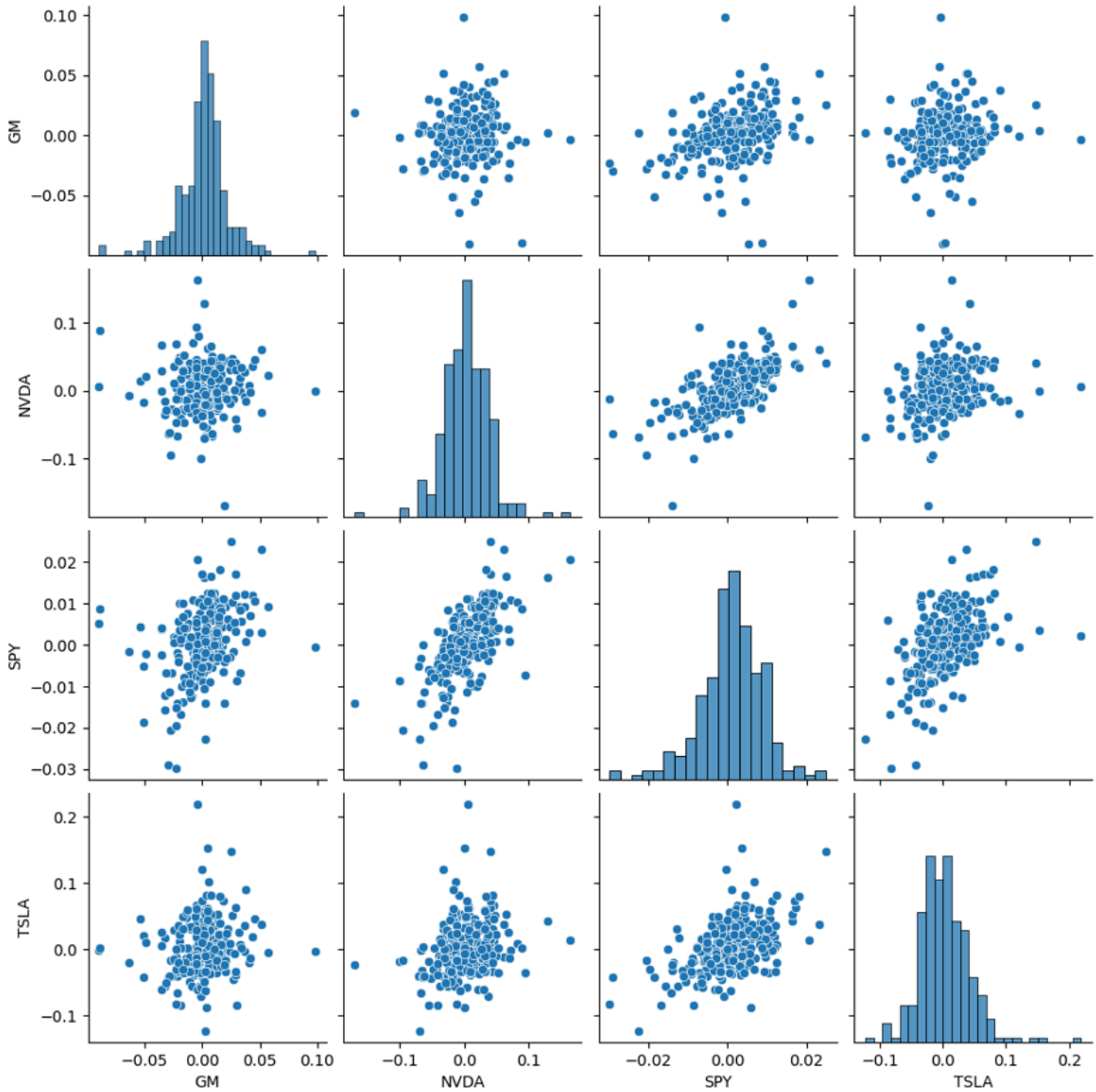
B. Daily Return



C. Distributions of Daily Return



D. Pair Plots for Correlation



Summary

A. Dataset Overview

The dataset utilized in this analysis consists of stock price data for four assets: SPY (S&P 500 ETF), TSLA (Tesla), GM (General Motors), and NVDA (NVIDIA). These stocks were chosen to

provide a diverse range of market exposure, spanning a broad-market ETF, an electric vehicle manufacturer, a legacy automotive company, and a high-growth technology firm. The dataset includes daily trading data over the past year, covering the period from February 2024 to February 2025, and captures key financial metrics such as Open, High, Low, and Close prices.

Since the stock market does not operate on weekends and certain public holidays, each stock has approximately 251 data points per year instead of 365. The data is structured as time series data, meaning it consists of stock prices recorded at regular intervals over time. This structure enables trend analysis, volatility assessment, and correlation studies between the selected stocks.

To analyze stock performance, I examined daily returns, which track the percentage change in closing prices from one trading day to the next. Volatility was measured using standard deviation, providing insights into the level of risk associated with each stock. To facilitate direct comparisons, I used a "Close 100" chart to normalize stock prices relative to their starting value, allowing for a clear visualization of how each stock performed over time in relation to one another and the broader market, represented by SPY.

Additionally, I analyzed the correlation between these stocks to understand how their price movements relate to each other. This helps identify potential diversification benefits and assess how different stocks respond to market trends. By understanding these relationships, more informed investment decisions can be made based on risk, performance, and market dynamics.

B. Graph Interpretation

Normalized Stock Price Index (Close 100):

This chart normalizes stock prices by setting them to a common starting value (100), making it easier to compare their relative performance over time. The data shows that NVDA and TSLA significantly outperformed SPY and GM. NVDA exhibited the strongest growth trajectory, reflecting substantial investor confidence and market trends favoring semiconductor stocks.

Tesla (TSLA), while also experiencing strong growth, showed more volatility with large price swings, indicating a higher-risk, high-reward investment profile. In contrast, GM and SPY demonstrated more stable, steady performance. SPY, representing the broader market, maintained a gradual upward trend, while GM, though underperforming NVDA and TSLA, remained relatively stable.

This visualization highlights the differences in risk and return across these stocks, helping investors assess which assets align with their investment goals.

Daily Return Plots:

The daily return graphs illustrate how much each stock's price changed from one day to the next, highlighting their volatility. NVDA and TSLA exhibit the largest swings, reinforcing their higher-risk profile with significant price fluctuations. This suggests that while these stocks offer strong growth potential, they also come with greater uncertainty.

In contrast, SPY and GM display smaller, more stable daily returns, making them relatively safer options for investors seeking steady growth without extreme day-to-day volatility. This comparison helps in assessing risk tolerance and investment strategy based on price stability and return consistency.

Distribution of Daily Returns:

The histogram distributions of daily returns highlight the frequency of different price movements for each stock. TSLA and NVDA exhibit a wider spread, indicating more extreme price swings and reinforcing their higher volatility. This suggests that while these stocks have the potential for significant gains, they also come with increased risk.

On the other hand, GM's return distribution is more concentrated, signaling greater price stability. SPY also shows a tighter range of returns, reflecting its role as a broad-market ETF with relatively lower volatility. These patterns emphasize the trade-off between high-risk, high-reward stocks and more stable investment options.

Pair Plots for Correlation:

The scatter plots reveal how stock returns interact with each other. A strong correlation between SPY and NVDA suggests that both stocks respond similarly to macroeconomic factors, such as interest rate changes and overall market sentiment. This means that holding both may not provide significant diversification benefits, as their movements are closely linked.

On the other hand, the lower correlation between TSLA and NVDA indicates that these stocks do not always move in the same direction. This suggests potential diversification benefits—if one stock underperforms, the other may not necessarily follow, helping to balance risk in a portfolio. Understanding these relationships can be crucial for constructing a well-balanced investment strategy.

C. Investment Strategy

Based on the analysis, a well-balanced investment strategy should consider a mix of high-growth, stable, and lower-risk stocks to maximize returns while managing volatility. The suggested portfolio includes NVIDIA (NVDA), Tesla (TSLA), S&P 500 (SPY), and General Motors (GM), each offering unique strengths that contribute to a diversified approach.

Growth-Focused Allocation – 40% NVDA

NVDA has demonstrated exceptional growth, benefiting from strong investor confidence and trends favoring semiconductor stocks. This makes it a key driver for capital appreciation in the portfolio. While NVDA's potential returns are high, its volatility requires investors to be comfortable with market fluctuations.

High-Risk, High-Reward Component – 30% TSLA

TSLA adds an element of high-risk, high-reward investment. Its price swings are more extreme, but it has historically delivered strong long-term gains. Including TSLA diversifies the portfolio within the high-growth sector while offering a counterbalance to NVDA's semiconductor-driven performance.

Market Stability & Broad Exposure – 20% SPY

SPY provides diversification by mirroring the overall market. This helps reduce volatility and offers consistent returns over time. Including SPY ensures that the portfolio has a foundation of stability, making it resilient during market downturns.

Defensive & Low-Volatility Anchor – 10% GM

GM is the most stable stock in the portfolio, contributing to risk reduction. Its steady returns and lower volatility help smooth out fluctuations from NVDA and TSLA, ensuring that the portfolio remains balanced, even in volatile market conditions.

Summary

This portfolio balances growth and risk by leveraging NVDA and TSLA for high-return potential while using SPY for market stability and GM for lower volatility. Investors can benefit from diversification across different industries while aligning with long-term financial goals.

This allocation ensures that the portfolio can capture upside potential while mitigating extreme volatility, making it a well-rounded choice for investors looking for strong returns with controlled risk.